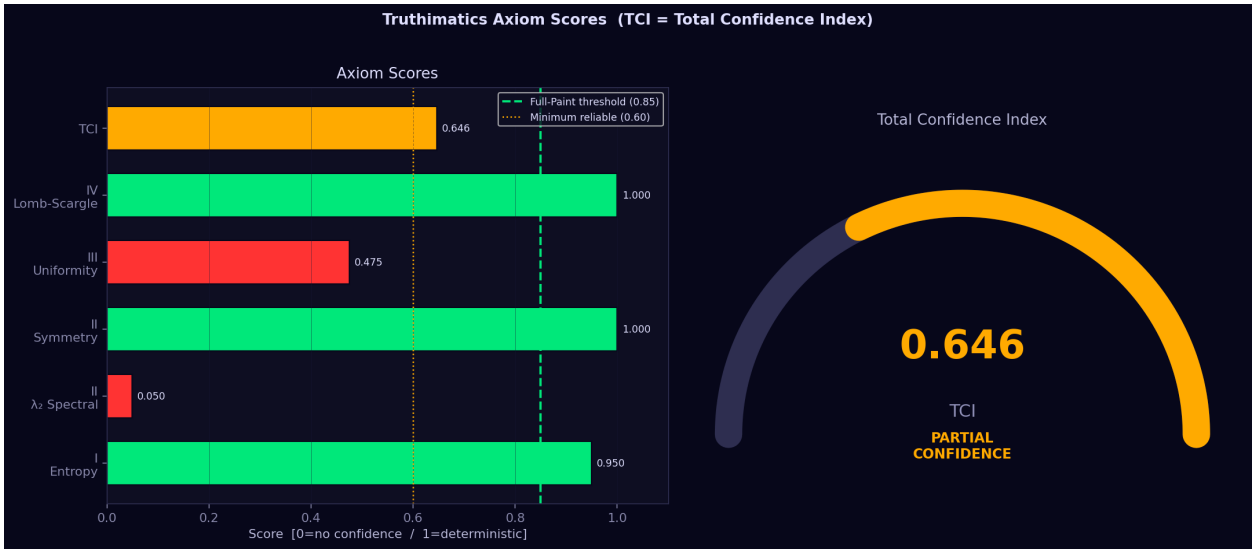
# AXIOM-DINOSAUR v4.0

Truthimatics Full-Paint Deterministic Synthesis

| | |
|---|---|
| **Specimen ID** | AXIOM_420DA50B9E |
| **Taxa** | Psittacosaurus |
| **Clade** | Ornithischian |
| **PBDB Records** | 70 |
| **Mass (regression)** | 3819.1 kg (Campione & Evans 2012) |
| **Mass (volumetric)** | 4065.3 kg (Silhouette integral, ρ=1060.0 kg/m³) |
| **Body Volume** | 4.279 m³ |
| **Validity Score** | 1.0000 / 1.0000 |
| **TCI** | 0.6462 (partial) |
| **Axiom I Entropy** | 0.9495 |
| **Axiom II λ■** | 0.1992 (symmetry 100%) |
| **Axiom III Uniformity** | 0.4748 |
| **Axiom IV Lomb FAP** | 1.000000 peak 0.000 Hz |
| **Integrity** | 68.0% Measured / 32.0% Inferred |
| **Elements** | 25 |
| **CoG (m)** | [0.287, 0.982, 0.0] |
| **Generated** | 2026-05-01 00:02 UTC |

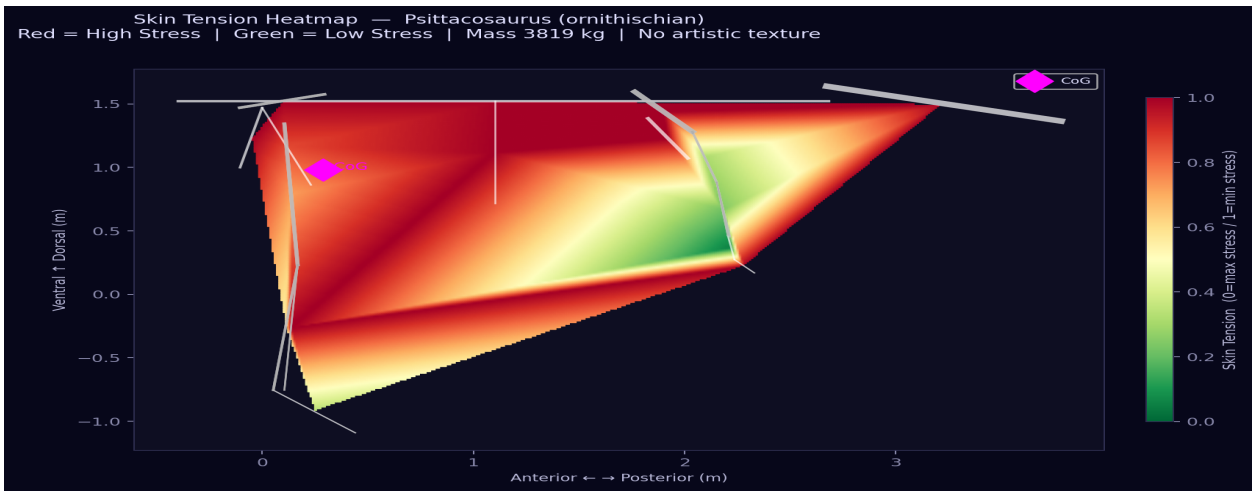
# 1 Truthimatics Confidence Analysis

TCI = weighted sum: Axiom I (entropy, w=0.25) + Axiom II (spectral, w=0.30) + Axiom III (Biewener uniformity, w=0.30) + Axiom IV (Lomb, w=0.15). TCI ≥ 0.85 authorises Full-Paint output.



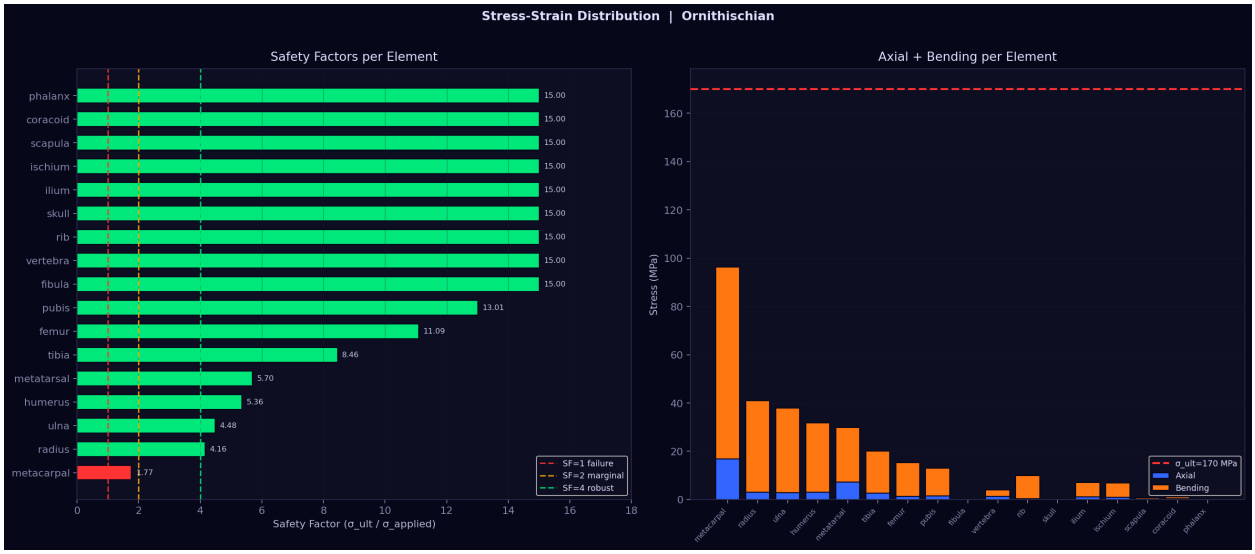
## 2 Skin Tension Heatmap

Stress field interpolated from per-bone safety factors onto the 2D body silhouette. Red=high stress (low SF), Green=low stress (high SF). No artistic texture applied.



# 3 Stress-Strain Distribution

Loading: quadruped (F=Mg/4). σ_ult=170 MPa (Reilly & Burstein 1975). SF≥4 robust.

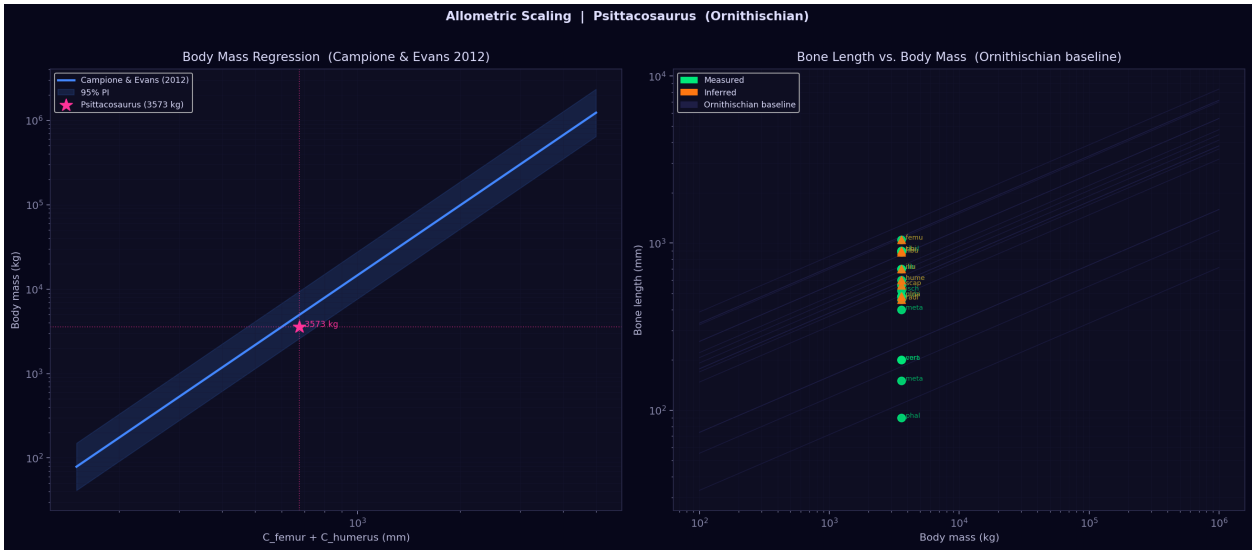



## Per-Element Stress Summary

| Element | σ_ax MPa | σ_bend MPa | σ_total MPa | SF | Status |
|---|---|---|---|---|---|
| coracoid | 0.391 | 0.749 | 1.140 | 15.000 | OK |
| femur | 1.294 | 14.034 | 15.328 | 11.091 | OK |
| fibula | 0.000 | 0.000 | 0.000 | 15.000 | OK |
| humerus | 3.028 | 28.689 | 31.716 | 5.360 | OK |
| ilium | 1.116 | 5.993 | 7.109 | 15.000 | OK |
| ischium | 0.945 | 5.865 | 6.810 | 15.000 | OK |
| metacarpal | 16.777 | 79.450 | 96.227 | 1.767 | CRITICAL |
| metatarsal | 7.232 | 22.591 | 29.823 | 5.700 | OK |
| phalanx | 0.000 | 0.000 | 0.000 | 15.000 | OK |
| pubis | 1.563 | 11.506 | 13.068 | 13.008 | OK |
| radius | 3.063 | 37.821 | 40.883 | 4.158 | OK |
| rib | 0.298 | 9.588 | 9.886 | 15.000 | OK |
| scapula | 0.158 | 0.541 | 0.699 | 15.000 | OK |
| skull | 0.000 | 0.000 | 0.000 | 15.000 | OK |
| tibia | 2.701 | 17.401 | 20.102 | 8.457 | OK |
| ulna | 2.832 | 35.084 | 37.916 | 4.484 | OK |
| vertebra | 1.294 | 2.673 | 3.968 | 15.000 | OK |

# 4 Allometric Scaling Analysis

Campione & Evans (2012): log■■(M_g)=2.754×log■■(C_f+C_h)–1.097 ($R^2$=0.97; N=426 non-avian dinosaurs).



## 5 Physical Properties Analysis

Biomechanical performance estimates derived deterministically from skeletal geometry. Speed: Alexander (1976) locomotion equation. Bite force: Christiansen & Wroe (2007). All values represent physically constrained estimates, not artistic approximations.

| Parameter | Value | Notes |
|---|---|---|
| **Hip Height** | 2.35 m | Femur + tibia + metatarsal |
| **Estimated Speed** | 14.6 km/h  (4.06 m/s) | Gait: Trot \| Fr=0.715 |
| **Stride Length** | 4.875 m | Alexander (1976) / Thulborn (1990) |
| **Bite Force** | 17493.1 N  (1783.2 kgf) | Ornithischian regression (Norman & Weishampel 1991) |
| **Skull Length** | 900.0 mm | Mid-shaft proxy from skull feature vector |
| **Femoral Robusticity** | 0.389 | Circumference / length (dimensionless) |
| **Limb Strength Index** | 4e-05 | (A_f + A_t) / M^0.67  (Christiansen 1998) |
| **Est. Body Length** | 9.43 m | Allometric from femur length |
| **Est. Neck Length** | 1.04 m | Allometric from femur length |
| **Est. Tail Length** | 2.64 m | Allometric from femur length |

## 6 Complete Bone Manifest

| Element | L (mm) | D (mm) | Vol cm³ | Mass g | Method | Conf | SF |
|---|---|---|---|---|---|---|---|
| coracoid | 200.0 | 140.0 | 1570.17 | 1960.0 | Measured | 0.88 | 15.00 |
| femur | 1050.0 | 130.0 | 7107.81 | 8872.7 | Measured | 0.88 | 11.09 |
| femur_right | 1050.0 | 130.0 | 7107.81 | 8872.7 | Symmetry | 0.84 | 11.09 |
| fibula | 880.0 | 40.0 | 398.10 | 417.5 | Measured | 0.88 | 15.00 |
| fibula_right | 880.0 | 40.0 | 398.10 | 417.5 | Symmetry | 0.84 | 15.00 |
| humerus | 600.0 | 85.0 | 1736.40 | 2167.5 | Measured | 0.88 | 5.36 |
| humerus_right | 600.0 | 85.0 | 1736.40 | 2167.5 | Symmetry | 0.84 | 5.36 |
| ilium | 700.0 | 140.0 | 5495.59 | 6860.1 | Measured | 0.88 | 15.00 |
| ilium_right | 700.0 | 140.0 | 5495.59 | 6860.1 | Symmetry | 0.84 | 15.00 |
| ischium | 520.0 | 90.0 | 1687.13 | 2106.0 | Measured | 0.88 | 15.00 |
| metacarpal | 150.0 | 35.0 | 27.42 | 22.6 | Measured | 0.88 | 1.77 |
| metatarsal | 400.0 | 55.0 | 484.67 | 605.0 | Measured | 0.88 | 5.70 |
| phalanx | 90.0 | 28.0 | 8.51 | 6.6 | Measured | 0.88 | 15.00 |
| pubis | 480.0 | 70.0 | 942.10 | 1176.0 | Measured | 0.88 | 13.01 |
| radius | 460.0 | 50.0 | 460.64 | 575.0 | Measured | 0.88 | 4.16 |
| radius_right | 460.0 | 50.0 | 460.64 | 575.0 | Symmetry | 0.84 | 4.16 |
| rib | 700.0 | 35.0 | 343.47 | 428.8 | Measured | 0.88 | 15.00 |
| scapula | 560.0 | 220.0 | 10856.59 | 13552.3 | Measured | 0.88 | 15.00 |
| scapula_right | 560.0 | 220.0 | 10856.59 | 13552.3 | Symmetry | 0.84 | 15.00 |
| skull | 900.0 | 400.0 | 57679.64 | 72001.5 | Measured | 0.88 | 15.00 |
| tibia | 900.0 | 90.0 | 2920.03 | 3645.1 | Measured | 0.88 | 8.46 |
| tibia_right | 900.0 | 90.0 | 2920.03 | 3645.1 | Symmetry | 0.84 | 8.46 |
| ulna | 480.0 | 52.0 | 519.89 | 649.0 | Measured | 0.88 | 4.48 |
| ulna_right | 480.0 | 52.0 | 519.89 | 649.0 | Symmetry | 0.84 | 4.48 |
| vertebra | 200.0 | 130.0 | 1353.87 | 1690.0 | Measured | 0.88 | 15.00 |